Ghost Steps

Invisible Energy Made Visible

An interactive art installation that transforms everyday footsteps into a visible energy ecosystem through piezoelectric technology

Declaration

We hereby declare that this report, titled "Ghost Steps – Invisible Energy Made Visible", is our own work and has not been submitted for any other module or degree. All sources of information used have been acknowledged appropriately in the text and in the bibliography.

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1. Introduction

Ghost Steps is an interactive art experience that transforms everyday footsteps into a visible energy ecosystem by harvesting piezoelectric energy from participants as they move through the space.

The installation reveals the usually invisible nature of electricity through light painting, projections, and an illuminated urban environment driven directly by human movement. Energy generated from footsteps powers a digital cityscape projection that visualizes how much energy is being produced by collective foot traffic.

Project Alignment

The project aligns with the **United Nations Sustainable Development Goal 7: Affordable and Clean Energy**, by demonstrating how accessible, renewable human kinetic energy can be repurposed as a meaningful resource.

Through storyboarding, system specification, and conceptual technology design, Ghost Steps is scoped as a **real-time cyber-physical installation** that merges sensing, computation, and audiovisual feedback into a cohesive interactive experience.

2. Concept Overview

Ghost Steps functions as an interactive installation in which users generate and visualize energy simply by walking through the environment. The installation is set in a dimly lit space featuring piezoelectric floor tiles, projection surfaces, and synchronized audio-visual output.

Ghost Trails

Each footstep on the floor tiles produces electrical energy, translated into glowing "ghost trails" that trace participants' movement paths across the ground.

Collective Energy

As more people interact with the surface, the total stored energy increases, demonstrating the power of collective action.

Virtual City

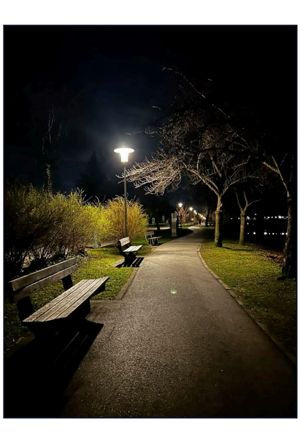
Collective energy powers a virtual city projected on a nearby wall that gradually transitions from dark and barren to brightly illuminated and lively.

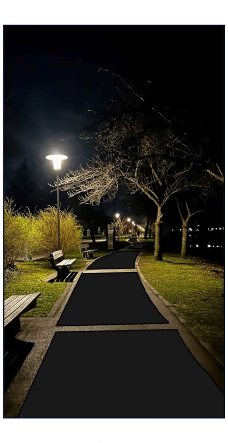
This setup emphasizes **cause-and-effect**: visitor movement becomes both the trigger and fuel for the visual transformation, making users aware that their bodies are acting as a **renewable energy source** for the environment.

3. Storyboard Description

The storyboard illustrates Ghost Steps through a park-based scenario that shows how the system behaves over time in four key frames.

3.1 Frame 1 – Idle Park Environment

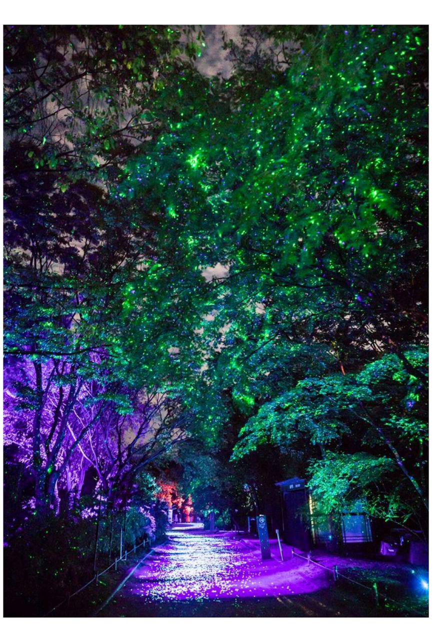




The park appears normal, with no visible interactive elements on the surface. Piezoelectric tiles are embedded beneath the walking path but remain inactive because no one is currently stepping on them. The system is in an idle state, passively waiting for footsteps to generate energy and activate the interactive installation.

3.2 Frame 2 – First Footsteps and Activation

When a visitor steps onto the piezoelectric path, kinetic energy is generated by the pressure of their footsteps. The system immediately responds by illuminating the tiles and visualizing the energy produced at each step. As more users join, multiple light activations appear across the path, representing collective energy generation and the beginning of system activation.



3.3 Frame 3 – Environment Fully Activated

As users continue to walk, the system accumulates kinetic energy and transforms the park into a fully active interactive space. Ghost steps manifest as glowing light trails along the path, while surrounding elements such as trees and ground illumination also respond dynamically to the generated energy. This visual feedback makes the invisible energy of human movement perceptible and tangible to participants.

3.4 Frame 4 – Energy Conversion and Functionality

The final frame emphasizes the underlying energy conversion principles. Piezoelectric tiles convert kinetic energy from footsteps into electrical energy using embedded pressure sensors. This electrical energy is transmitted to the system, where it powers lighting, wayfinding, signage, wireless communication elements, phone charging, and other interactive components, demonstrating how human movement can sustain an interactive environment.

4. Conceptual Technology

Ghost Steps is organized as a **real-time cyber-physical interaction system** that integrates sensing hardware, computational processing, and audiovisual output into a continuous feedback loop.

4.1 Input Layer – Sensing

- Piezoelectric transducers or FSRs
- Integrated into modular floor tiles
- Detect pressure from footsteps
- Convert mechanical stress to voltage
- Microcontroller (Arduino/ESP32) samples, filters, and digitizes data

4.2 Processing Layer – Data Interpretation

- Data sent via Serial or OSC
- Real-time media engine (Unity/TouchDesigner)
- Normalize and map sensor data
- Drive particle systems (ghost trails)
- Shader-based lighting for cityscape
- Global energy variable tracking

4.3 Output Layer – Audiovisual Feedback

- High-resolution projector
- Addressable LED floor lighting
- Spatialized audio speakers
- Synchronized multimodal feedback
- Visible & audible consequences
- Renewed user engagement

Feedback Loop

Overall, the system forms a closed feedback loop:

Physical Action → Sensor Transduction → Computation → A/V Response → User Engagement

5. Interaction Rules and Algorithms

The interaction logic for Ghost Steps is governed by an **energy aggregation model** combined with **temporal decay**, which helps communicate ideas of sustainability and continuous input.

5.1 Energy Mapping

Each sensor continuously monitors pressure values from the floor. When the pressure exceeds a defined threshold T, a kinetic energy unit E is computed using a normalization function:

```
E = map(pressure_value, min_input, max_input, 0, max_energy)
```

This energy E is added to a global state variable storing the total energy:

```
total_energy = total_energy + E
```

5.2 Visual and System Mapping

The global energy value is mapped to different visual and systemic behaviors using proportional relationships:

- Ghost Trail Opacity: Proportional to instantaneous energy input
- Trail Persistence: Proportional to magnitude of energy contributed
- City Illumination: Proportional to cumulative total_energy
- Activated Buildings: Increases as energy thresholds are crossed

These mappings ensure that individual footsteps contribute to a shared visual result, reinforcing the notion of **collective impact**.

5.3 Temporal Decay

To simulate real-world energy consumption, a decay function is applied when no input is detected for a specified time interval.

```
if no_input_duration > t:
 total_energy = total_energy - decay_rate
```

This temporal dependency emphasizes that the environment only remains bright and active through **continuous collective participation**.

6. System Flow

The system operates as a **real-time interaction pipeline**, moving from detection to feedback and decay in a continuous loop.

Initial State

The virtual city is projected at low luminosity, representing an energy-deficient environment waiting to be powered by human movement.

Phase 1 – Detection

When a participant steps onto the interactive surface, embedded piezoelectric sensors detect mechanical deformation and produce electrical signals.

Phase 2 – Signal Acquisition and Processing

The microcontroller samples the analog signal, filters noise, applies thresholds, and converts it into calibrated digital energy units.

Phase 3 – Data Transmission

These energy units are transmitted to the visual processing engine via serial communication or OSC.

Phase 4 – Computational Mapping

The software updates the global energy variable and maps it to visual and audio parameters using scaling algorithms.

Phase 5 – Visual and Auditory Output

Luminous particle paths (ghost trails) appear where the participant steps, and the projected city progressively illuminates based on accumulated energy, while background audio becomes richer and more layered.

Phase 6 – Feedback and Decay

When interaction stops, a programmed decay function reduces the stored energy, leading to a gradual dimming of the city and a softening of the audiovisual output.

This cycle loops continuously, making Ghost Steps a **living system** that depends on ongoing human input to sustain its virtual urban environment.

7. Theme Exploration – UN SDG 7

Ghost Steps directly engages with the theme of sustainable energy, particularly focusing on **SDG 7: Affordable and Clean Energy**. The installation emphasizes that human kinetic energy—typically wasted in everyday activities—can be harnessed and repurposed into useful electrical energy.

UN Sustainable Development Goal 7

Affordable and Clean Energy

Energy Access

Ensure universal access to affordable, reliable and modern energy services

Renewable Share

Increase substantially the share of renewable energy in the global energy mix

Energy Efficiency

Double the global rate of improvement in energy efficiency

Clean Technology

Enhance international cooperation to facilitate access to clean energy research and technology

Human-Powered Energy: Demonstrates how everyday human movement can be converted into usable electrical energy

Urban Integration: Illustrates potential for integrating renewable human energy into future urban infrastructures

Behavioral Awareness: Encourages participants to reflect on their role in energy systems and sustainable practices

By visualizing a city that only comes to life through collective footstep energy, Ghost Steps demonstrates both a **literal and metaphorical connection** between human behavior and energy systems. It highlights the potential of alternative energy approaches and reinforces the idea that future urban environments could incorporate human-powered, renewable solutions as part of their energy mix.

8. Artistic Intent

The artistic intent behind Ghost Steps is to re-imagine invisible energy as an engaging, playful, and thought-provoking interactive experience. The installation aims to examine the relationship between humans, technology, and sustainability, not by prescribing specific behaviors, but by encouraging participants to reflect on how they generate and consume energy in daily life.

Making the Invisible Visible: Transform invisible energy flows into tangible, visual experiences that participants can see, understand, and interact with in real time.

Playful Exploration: Create an engaging, playful environment that encourages curiosity and experimentation with energy generation and consumption.

Thought-Provoking Questions: Raise questions about our relationship with technology, sustainability, and energy systems without prescribing specific behaviors.

Aesthetic Experience: Deliver an immersive, aesthetically rich experience that goes beyond didactic messaging to create emotional resonance.

By making participants responsible for powering a city, Ghost Steps prompts them to consider their role in broader energy systems and to imagine how human-led renewable energy could shape future urban infrastructures. In this sense, the work serves both as a **speculative design for energy-conscious cities** and as an **invitation to rethink personal and collective energy habits**.

9. Reflection and Learning Outcomes

Developing Ghost Steps helped clarify how interactive design, sustainability, and systems thinking can intersect in a single project. Storyboarding and technical specification made it easier to imagine how users would engage with the installation and how the system would respond in real time.

Storyboarding & Technical Specification: Storyboarding and technical specification made it easier to imagine how users would engage with the installation and how the system would respond in real time.

Cause-and-Effect Relationships: The project highlighted how every user action has a direct consequence in an interactive system, reinforcing the importance of clear cause-and-effect relationships in experience design.

Collaborative Development: Collaboration among contributors enriched the concept, showing how diverse perspectives lead to more meaningful and nuanced interactive artworks.

Systems Thinking: Understanding how sensing, processing, and output layers work together demonstrated the complexity and beauty of integrated cyber-physical systems.

Overall, Ghost Steps stands as a strong example of how **interactive media can promote sustainable thinking** while offering an **immersive, aesthetically rich experience**. The project demonstrated that meaningful interactive art can emerge from the intersection of technical innovation, environmental consciousness, and creative expression.

10. Bibliography

- Interactive Media – System Specification Document (Ghost Steps – Invisible Energy Made Visible).
- Interactive Media – Concept and Reflection Document (Ghost Steps – Harvesting Humanity's Invisible Power).
- Interactive Media – Storyboarding PDF (Ghost Steps Park Scenario Frames).